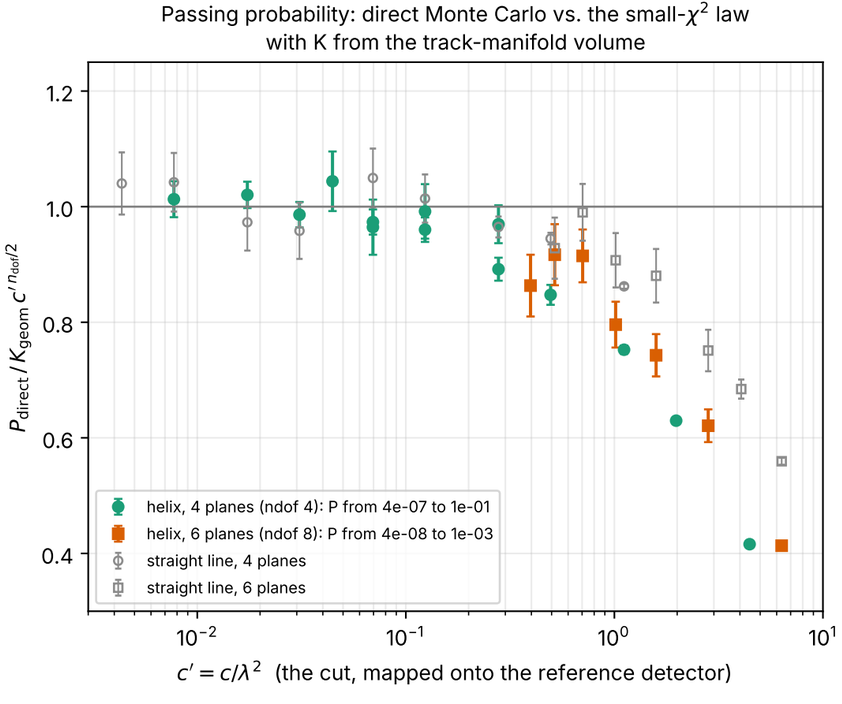
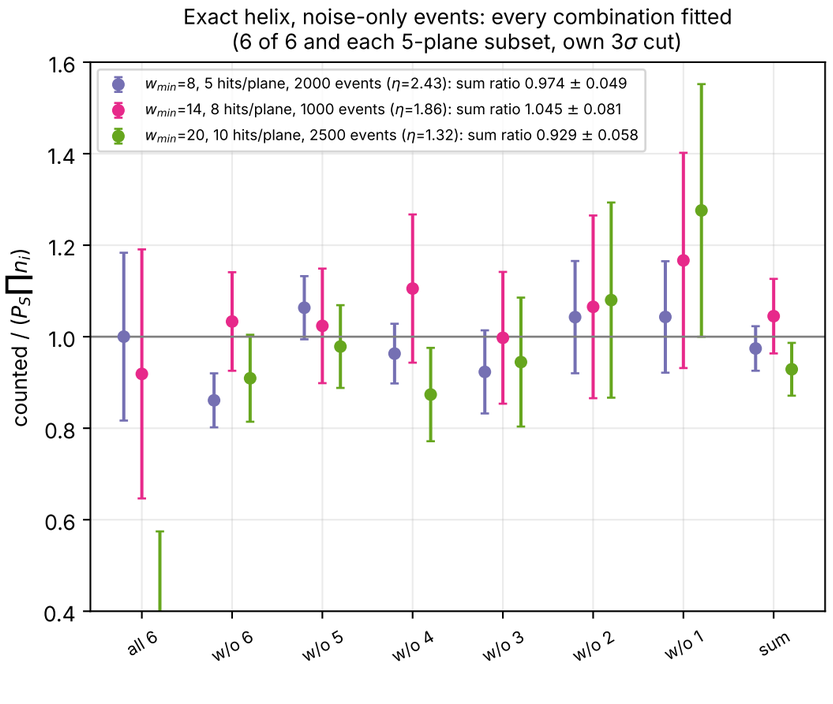
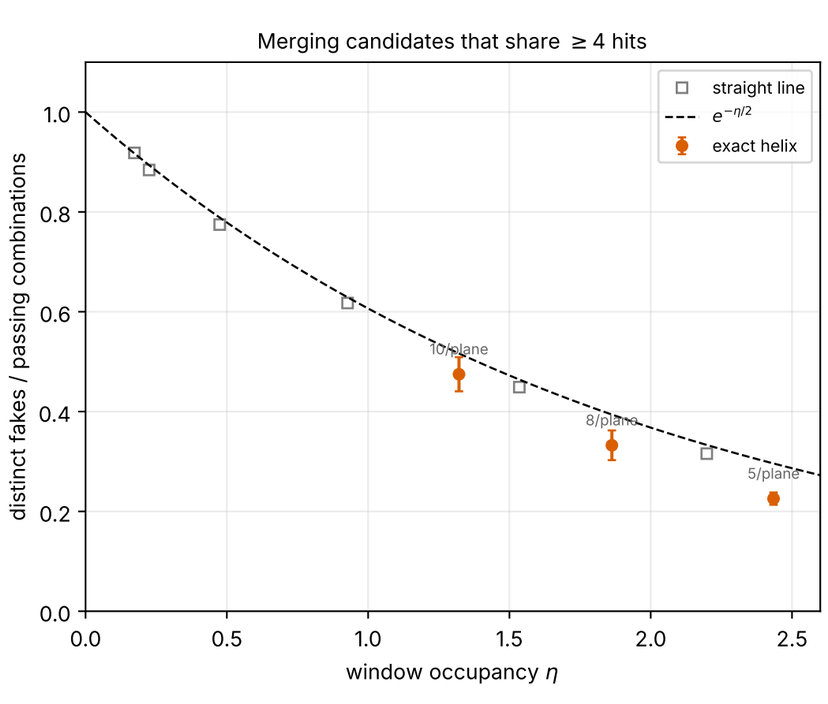

Passing probability: direct Monte Carlo vs. the small-χ^2 law
with K from the track-manifold volume
$P_{\mathrm{direct}} / K_{\mathrm{geom}}\, c'^{\,n_{\mathrm{dof}}/2}$
1.2
1.0
0.8
0.6
0.4
10^{-2}
10^{-1}
10^{0}
10^{1}
$c' = c/\lambda^2$ (the cut, mapped onto the reference detector)
helix, 4 planes (ndof 4): P from 4e-07 to 1e-01
helix, 6 planes (ndof 8): P from 4e-08 to 1e-03
straight line, 4 planes
straight line, 6 planes


Exact helix, noise-only events: every combination fitted
(6 of 6 and each 5-plane subset, own 3σ cut)
counted / $(P_S \prod n_i)$
1.6
1.4
1.2
1.0
0.8
0.6
0.4
w_{min}=8, 5 hits/plane, 2000 events (η=2.43): sum ratio 0.974 ± 0.049
w_{min}=14, 8 hits/plane, 1000 events (η=1.86): sum ratio 1.045 ± 0.081
w_{min}=20, 10 hits/plane, 2500 events (η=1.32): sum ratio 0.929 ± 0.058
all 6
w/o 6
w/o 5
w/o 4
w/o 3
w/o 2
w/o 1
sum


Merging candidates that share ≥ 4 hits
distinct fakes / passing combinations
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
window occupancy η
straight line
$e^{-\eta/2}$
exact helix
10/plane
8/plane
5/plane